

Computer Aided Geometric Design

— Bézier Curves

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Bézier Curves

Named after their inventor **Pierre Bézier**, an engineer with the Renault car company.

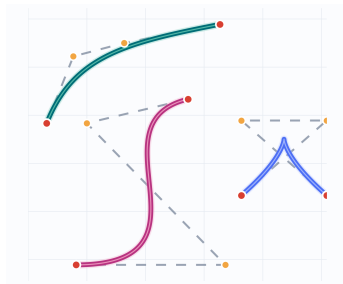


Figure: Cubic Bézier curves

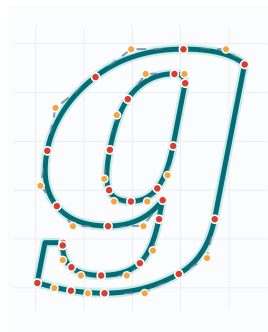


Figure: Font definition using Bézier curves

The Equation of a Bézier Curve

- ▶ Parameter $t \in [0, 1] \subset \mathbb{R}$
- ▶ Bernstein polynomials

$$B_i^n(t) := \binom{n}{i} (1-t)^{n-i} t^i, \quad i = 0, \dots, n$$

- ▶ Control points:

$$\mathbf{P}_0, \dots, \mathbf{P}_n \in \mathbb{R}^2$$

- ▶ Parameterization:

$$\mathbf{P}(t) := \sum_{i=0}^n \mathbf{P}_i B_i^n(t)$$

- ▶ Degree n Bézier curve
- ▶ Observe that $\mathbf{P}(0) = \mathbf{P}_0$, $\mathbf{P}(1) = \mathbf{P}_n$.

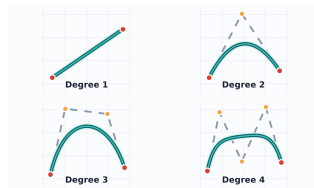


Figure: Bézier curves of various degree

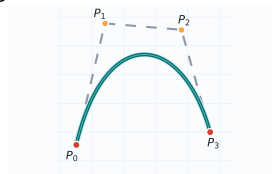


Figure: Control points of a cubic Bézier curve

Arbitrary Parameter Intervals

- ▶ In the previous definition, the Bézier curve starts at $t = 0$ and ends at $t = 1$.
- ▶ It is useful, especially for fitting together several Bézier curves, to allow an arbitrary parameter interval

$$t \in [t_0, t_1] \subset \mathbb{R}$$

such that $\mathbf{P}(t_0) = \mathbf{P}_0$ and $\mathbf{P}(t_1) = \mathbf{P}_n$.

- ▶ The modified parameterization is given by

$$\mathbf{P}_{[t_0, t_1]}(t) = \sum_{i=0}^n \mathbf{P}_i \binom{n}{i} \left(\frac{t_1 - t}{t_1 - t_0} \right)^{n-i} \left(\frac{t - t_0}{t_1 - t_0} \right)^i.$$

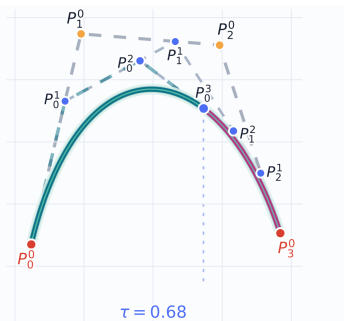
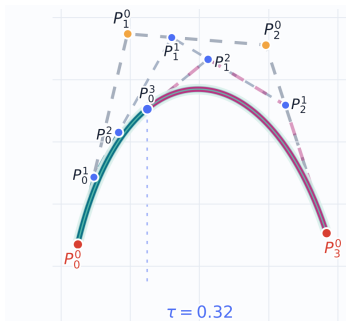
- ▶ It is obtained by a change of parameter: $t \leftarrow (t - t_0)/(t_1 - t_0)$.

The de Casteljau Algorithm

Subdivision

How to subdivide a Bézier curve $\mathbf{P}_{[t_0, t_2]}$ into two segments $\mathbf{P}_{[t_0, t_1]}$ and $\mathbf{P}_{[t_1, t_2]}$ whose union is equivalent to $\mathbf{P}_{[t_0, t_2]}$?

- Choose the subdivision parameter $\tau := \frac{t_1 - t_0}{t_2 - t_0}$.
- Label the control points as $\mathbf{P}_0^0, \dots, \mathbf{P}_n^0$.



The de Casteljau Algorithm

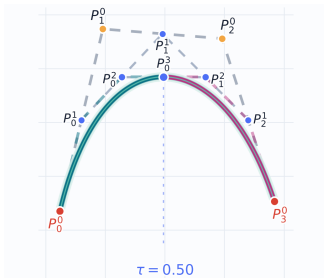
Recursive construction

For $j = 1, \dots, n$ and $i = 0, \dots, n - j$, compute

$$\mathbf{P}_i^j = (1 - \tau)\mathbf{P}_i^{j-1} + \tau\mathbf{P}_{i+1}^{j-1}.$$

Each new point is an affine interpolation between two consecutive points from the previous level.

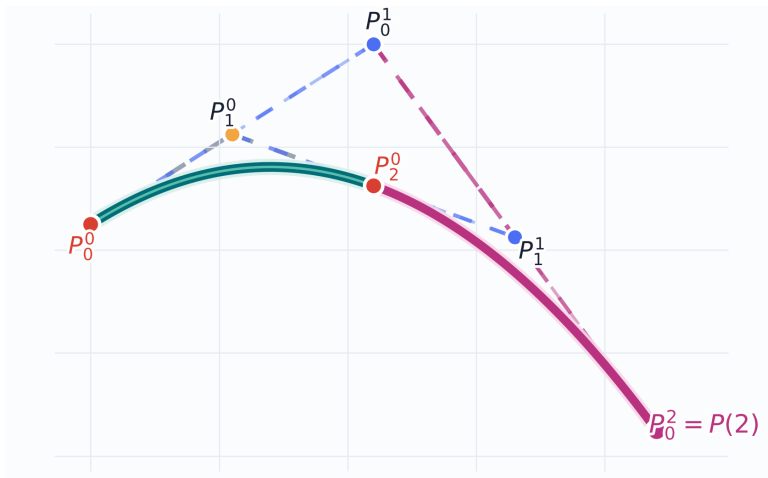
- ▶ Control points of $\mathbf{P}_{[t_0, t_1]}$: $\mathbf{P}_0^0, \mathbf{P}_0^1, \dots, \mathbf{P}_0^n$.
- ▶ Control points of $\mathbf{P}_{[t_1, t_2]}$: $\mathbf{P}_0^n, \mathbf{P}_1^{n-1}, \dots, \mathbf{P}_n^0$.
- ▶ Evaluation: $\mathbf{P}(t_1) = \mathbf{P}_0^n$.



The de Casteljau Algorithm

De Casteljau construction for $\tau = 2 \notin [0, 1]$:

- The algorithm still works, but extrapolation is not numerically stable.



The de Casteljau Algorithm

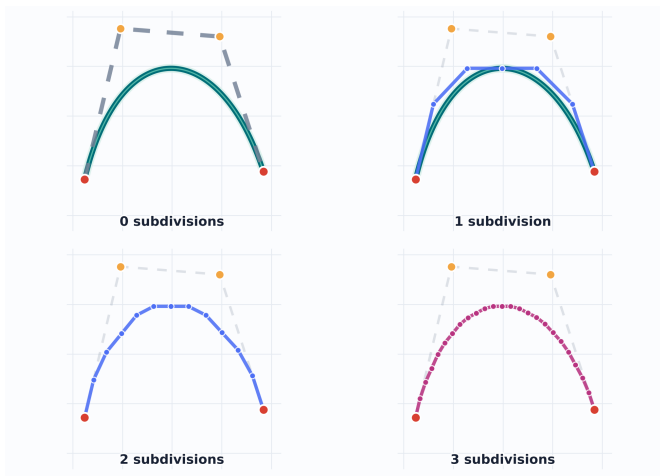


Figure: The collection of control polygons converges to the curve after repeated subdivisions.

New control points: $\mathbf{P}_0^* = \mathbf{P}_0, \quad \mathbf{P}_{n+1}^* = \mathbf{P}_n,$
 $\mathbf{P}_i^* = \frac{i}{n+1}\mathbf{P}_{i-1} + \frac{n+1-i}{n+1}\mathbf{P}_i, \quad i = 1, \dots, n.$

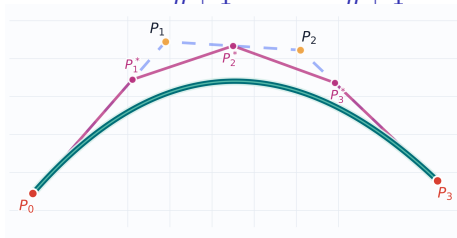


Figure: Degree elevation of a cubic Bézier curve

$$\mathbf{P}(t) = ((1-t) + t) \mathbf{P}(t) = ((1-t) + t) \sum_{i=0}^n \mathbf{P}_i B_i^n(t) = \sum_{i=0}^{n+1} \mathbf{P}_i^* B_i^{n+1}(t)$$

Degree Elevation

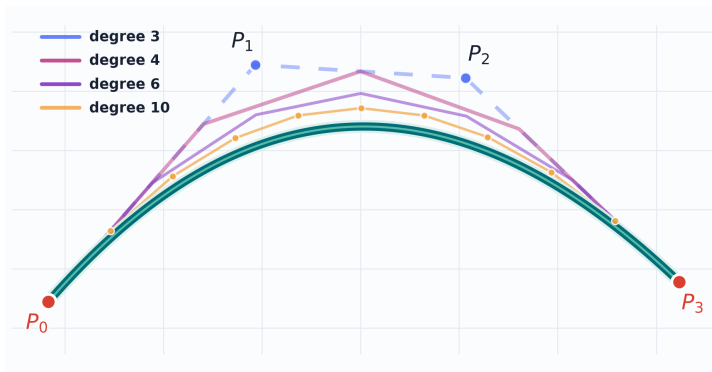


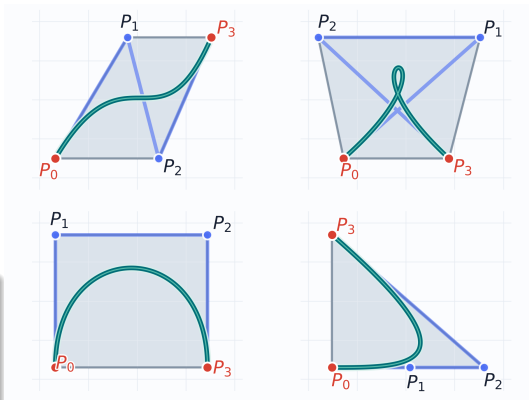
Figure: Repeated degree elevation converges to the curve

The Convex Hull Property of Bézier Curves

- ▶ Bézier curves always lie in the **convex hull** of their control points.
- ▶ The **convex hull** is the smallest **convex set** that contains the control points.
- ▶ This is an easy consequence of the definition of a Bézier curve.

Convex set

A convex set is a set of points such that, given any two points A, B in that set, the segment $[AB]$ joining them lies entirely within that set.

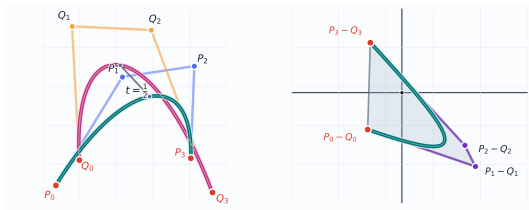


Distance between Two Bézier Curves

- **Difference curve:** Given two Bézier curves $\mathbf{P}(t) = \sum_{i=0}^n \mathbf{P}_i B_i^n(t)$ and $\mathbf{Q}(t) = \sum_{i=0}^n \mathbf{Q}_i B_i^n(t)$, define

$$\mathbf{D}(t) = \mathbf{P}(t) - \mathbf{Q}(t) = \sum_{i=0}^n (\mathbf{P}_i - \mathbf{Q}_i) B_i^n(t).$$

- The convex hull property implies that the distance between the two curves is bounded by the largest distance from the origin to any of the control points of $\mathbf{D}(t)$.



- **Attractive**
because easy to compute
- **Not very tight**
because it depends on the parameterizations

Variation Diminishing Property

Property

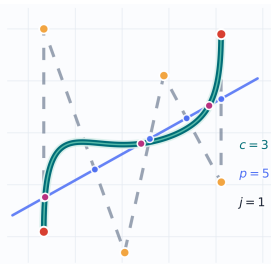
If a straight line intersects a Bézier curve in c points and the control polygon in p points, then it will always hold that

$$c = p - 2j$$

where j is zero or a positive integer.

Practical interpretation

A Bézier curve will "wiggle" no more than its control polygon.



First derivative – Hodograph

Given a Bézier curve $\mathbf{P}_{[t_0, t_1]}(t) = \sum_{i=0}^n \mathbf{P}_i B_i^n(t)$, its first derivative can be expressed as a Bézier curve $\mathbf{P}_{[t_0, t_1]}'(t) = \sum_{i=0}^{n-1} \mathbf{D}_i B_i^{n-1}(t)$ where

$$\mathbf{D}_i = \frac{n}{t_1 - t_0} (\mathbf{P}_{i+1} - \mathbf{P}_i).$$

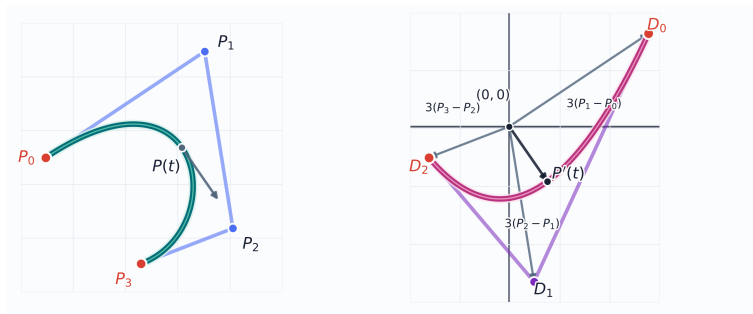


Figure: Hodograph Bézier curve, $[t_0, t_1] = [0, 1]$

Higher Derivatives

- ▶ The first derivative curve is known as a **hodograph**
- ▶ The second derivative can be computed as the hodograph of the hodograph, and similarly for the higher derivatives.
- ▶ It is convenient to compute them in **tabular form**.

Example

Let $\mathbf{P}_{[0, \frac{1}{2}]}(t)$ be a cubic Bézier curve with the control points

$$\mathbf{P}_0 = (2, 1), \quad \mathbf{P}_1 = (4, 5), \quad \mathbf{P}_2 = (8, 6), \quad \mathbf{P}_3 = (9, 2).$$

Then, we get the following hodographs

Curve	Control Points			
$\mathbf{P}(t)$	(2, 1)	(4, 5)	(8, 6)	(9, 2)
$\mathbf{P}'(t)$	(12, 24)	(24, 6)	(6, -24)	
$\mathbf{P}''(t)$	(48, -72)	(-72, -120)		
$\mathbf{P}'''(t)$	(-240, -96)			

Three Dimensional Bézier Curves

- ▶ Bézier control points are defined in the 3D space
- ▶ The resulting curve is hence a 3D curve
- ▶ **All the previous discussion extends to 3D without modification**

Remark : conic Bézier curves are always planar. This is because they are defined by only three control points.

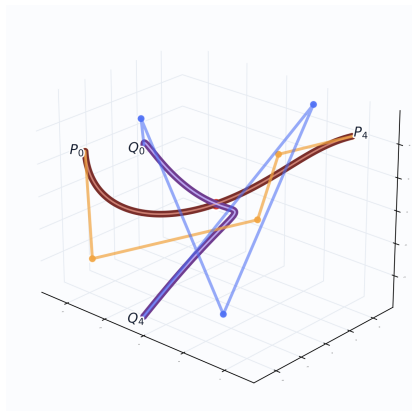


Figure: Intersection of two quartic Bézier curves in space

Rational Bézier Curves

- ▶ Each control point $\mathbf{P}_i$ is assigned a **scalar weight** w_i .
- ▶ The equation of a **rational Bézier curve** of degree n is

$$\mathbf{P}(t) = \frac{\sum_{i=0}^n w_i \mathbf{P}_i B_i^n(t)}{\sum_{i=0}^n w_i B_i^n(t)}$$

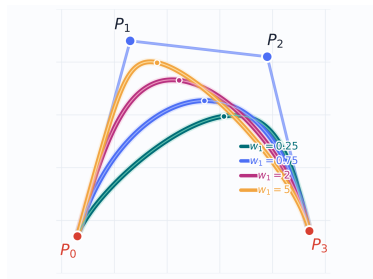


Figure: Impact of the weights

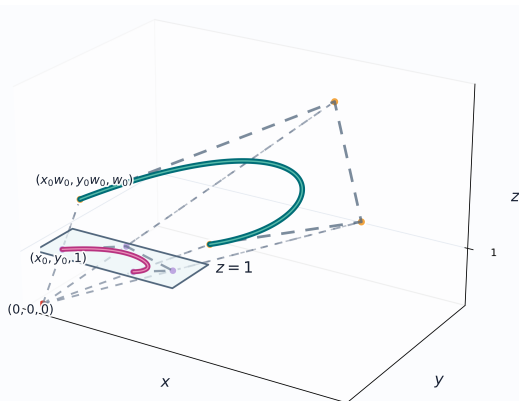
- ▶ "Rational" because the parameterization is given by rational functions (ratio of polynomials)
- ▶ Provides more control on the shape
- ▶ Allows exact representation of conic sections
- ▶ If all $w_i = 1$ then rational Bézier curves reduce to (polynomial) Bézier curves

Rational Bézier Curves

A rational Bézier curve can be interpreted as the perspective (or central) projection of a 3D Bézier curve.

- ▶ The 3D curve is a Bézier curve
- ▶ The 2D curve in the plane $z = 1$ is the perspective of the 3D curve: a rational Bézier curve.
- ▶ If $(x(t), y(t), z(t))$ denotes the points on the 3D curve, then the points on the 2D curve are given by

$$\left(\frac{x(t)}{z(t)}, \frac{y(t)}{z(t)}, 1 \right).$$



De Casteljau Algorithm and Degree Elevation

Extension to rational Bézier curves

Both the de Casteljau and the degree elevation algorithms extend easily to rational curves

- ▶ First, convert the rational Bézier curve into its corresponding 3D curve,
- ▶ Next, perform the algorithm on this 3D curve
- ▶ Finally, map the result back to 2D

Caution

The above procedure does not apply for hodographs because of the quotient rule differentiation.

- ▶ Nevertheless, we will describe how to compute the first derivative at an endpoint, as well as the curvature
- ▶ Used with the de Casteljau algorithm, this allows to compute the first derivative, or the curvature, at any point on a rational Bézier curve.

First derivative at an endpoint

For a rational Bézier curve of degree n

$$\mathbf{P}_{[t_0, t_1]}(t) = \frac{\sum_{i=0}^n w_i \mathbf{P}_i \binom{n}{i} \left(\frac{t_1-t}{t_1-t_0}\right)^{n-i} \left(\frac{t-t_0}{t_1-t_0}\right)^i}{\sum_{i=0}^n w_i \binom{n}{i} \left(\frac{t_1-t}{t_1-t_0}\right)^{n-i} \left(\frac{t-t_0}{t_1-t_0}\right)^i},$$

the first derivative at $t = t_0$ is

$$\mathbf{P}'(t_0) = \frac{w_1}{w_0} \frac{n}{t_1 - t_0} (\mathbf{P}_1 - \mathbf{P}_0)$$

and the second derivative at $t = t_0$ is

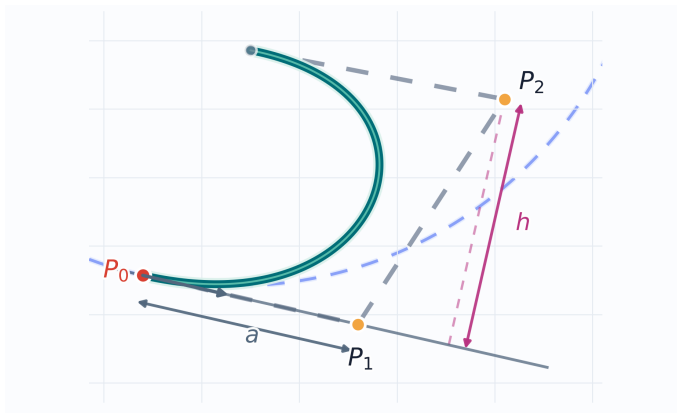
$$\mathbf{P}''(t_0) = \frac{n(n-1)}{(t_1 - t_0)^2} \frac{w_2}{w_0} (\mathbf{P}_2 - \mathbf{P}_0) - \frac{2n}{(t_1 - t_0)^2} \frac{w_1}{w_0} \frac{nw_1 - w_0}{w_0} (\mathbf{P}_1 - \mathbf{P}_0).$$

Remark: One just applies the quotient rule differentiation and evaluates at $t = t_0$; computations quickly become extremely complicated.

Curvature at an endpoint

- ▶ Rational Bézier curve of degree n
- ▶ a and h as on the picture

$$\kappa(t_0) = \frac{w_0 w_2}{w_1^2} \frac{n-1}{n} \frac{h}{a^2}$$



Continuity

Definition

Two curve segments $\mathbf{P}_{[t_0, t_1]}$ and $\mathbf{Q}_{[t_1, t_2]}$ are said to be $\mathcal{C}^k$ continuous if

$$\mathbf{P}(t_1) = \mathbf{Q}(t_1), \mathbf{P}'(t_1) = \mathbf{Q}'(t_1), \dots, \mathbf{P}^{(k)}(t_1) = \mathbf{Q}^{(k)}(t_1).$$

For example, these two curves are

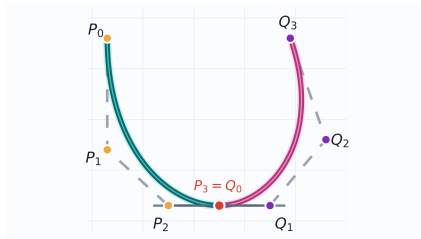
- ▶ $\mathcal{C}^0$ if $\mathbf{P}_3 = \mathbf{Q}_0$,
- ▶ $\mathcal{C}^1$ if in addition

$$\frac{3}{t_1 - t_0}(\mathbf{P}_3 - \mathbf{P}_2) = \frac{3}{t_2 - t_1}(\mathbf{Q}_1 - \mathbf{Q}_0)$$

- ▶ $\mathcal{C}^2$ if in addition

$$\frac{6}{(t_1 - t_0)^2}(\mathbf{P}_3 - 2\mathbf{P}_2 + \mathbf{P}_1) = \frac{6}{(t_2 - t_1)^2}(\mathbf{Q}_2 - 2\mathbf{Q}_1 + \mathbf{Q}_0)$$

- ▶ etc.



Geometric Continuity

- ▶ The **geometric continuity**, denoted G^k , is another method for describing the continuity of two curves, that is **independent of their parameterizations**.
- ▶ The conditions for geometric continuity are less strict compared to the classical continuity.

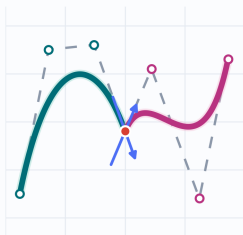
Definition

Two curves are G^k at a given point P if they can be reparameterized so that they become C^k at this point P .

- ▶ G^0 : the two curves have a common endpoint, but not necessarily with the same parameter value,
- ▶ G^1 : the line segments $\mathbf{P}_2 - \mathbf{P}_3$ and $\mathbf{Q}_0 - \mathbf{Q}_1$ are collinear (i.e. common tangent line),
- ▶ G^2 : same tangent line and same center of curvature,
- ▶ etc.

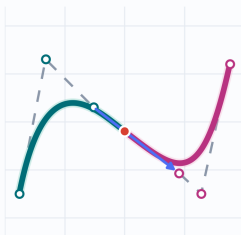
Geometric Continuity: Examples

G^0



common endpoint

G^1 but not C^1



common tangent, different speeds

G^2 but not C^2



same tangent but different accelerations